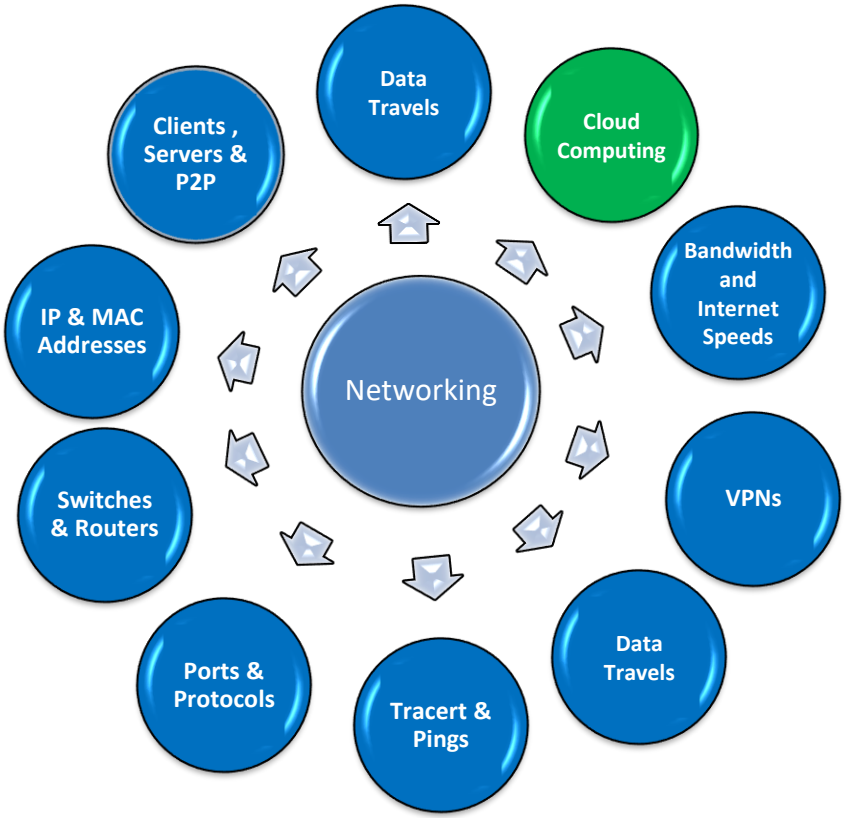
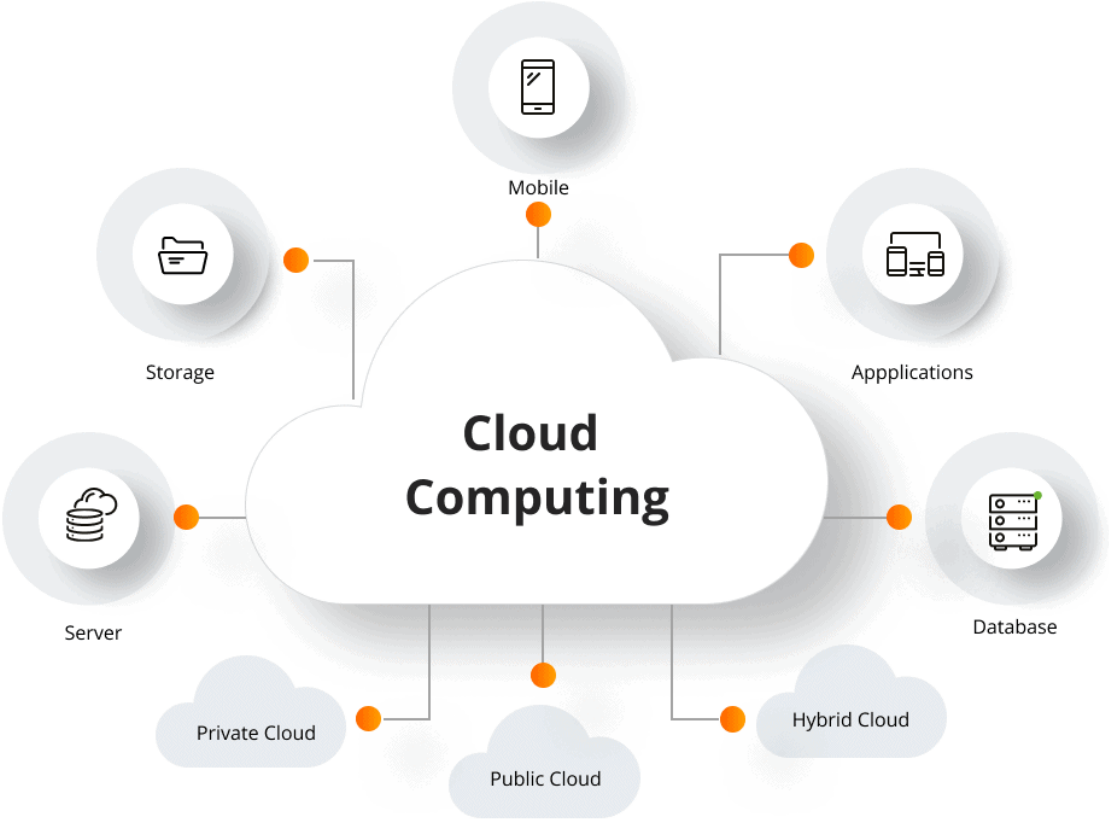

Networking
Data Travels
Cloud Computing
Bandwidth and Internet Speeds
VPNs
Data Travels
Tracert & Pings
Ports & Protocols
Switches & Routers
IP & MAC Addresses
Clients , Servers & P2P

# Cloud Computing



# What is Cloud Computing?

•**Definition:**
Cloud computing means using **Internet-based servers** to store data, run software, and manage services — instead of keeping everything on your local computer.

In simple words:

“You use someone else’s computer (over the internet) to store and process your stuff.”

Examples:

☁ Google Drive ☁ iCloud ☁ Microsoft 365 ☁ Adobe Cloud

# How It Works

1. Your device (phone, laptop) connects to the **Internet**.

2. You send a request to a **cloud server** (for example: upload photo to Instagram).

3. The server **stores or processes** your data.

4. You can access the same data from **any device, anywhere**.

- ❑ **No local storage needed**, only an internet connection.

# Why It’s Important

- ✅ Access from anywhere
- ✅ No need for physical servers
- ✅ Automatic software updates
- ✅ Collaboration and sharing made easy
- ✅ Scalable — increase or decrease storage anytime

**Example:**
Upload photos from your phone → open them later on a laptop → same files appear instantly.

# Real-World Examples

| Company | Cloud Service | Purpose |
|---|---|---|
| Google Drive | File storage | Access docs anywhere |
| Microsoft 365 (Office Online) | SaaS | Use Word & Excel online |
| Adobe Creative Cloud | SaaS | Edit photos/videos via the web |
| Amazon AWS | IaaS | Rent servers for businesses |
| Heroku / Azure | PaaS | Build and deploy apps |

# Four Main Types of Cloud Services

1. **SaaS (Software as a Service)**
- Use ready-made applications online.
- Examples: Gmail, Canva, Slack, Instagram.
- ✅ No installation needed.
2. **PaaS (Platform as a Service)**
- Rent tools to build your own apps.
- Example: Google App Engine, Microsoft Azure.
- 🧰 Developers focus on coding, not hardware.
3. **IaaS (Infrastructure as a Service)**
- Rent servers and storage.
- Example: Amazon AWS, Google Cloud Platform.
- 🏗️ Build apps using your own tools.
4. **FaaS (Function as a Service)**
- Pay only for specific functions you use (e.g., event-based).
- Example: AWS Lambda.
- ⏱️ Cost-efficient for small tasks.

# Advantages of Cloud Computing

- Access from anywhere, on any device
- Data backup and recovery
- Cost-effective — pay only for what you use
- Scalability — add storage easily
- Supports teamwork and remote work

# Disadvantages / Challenges

- Requires Internet connection
- Potential data privacy risks
- Dependent on third-party providers
- May face downtime or service outages

# Section Quiz

**Question 1:**

What does MAC stand for?

A) Machine Access Control
B) Media Access Control
C) Multiple Access Control
D) Media Access Consol

**Question 2:**

Which device is used to connect multiple computers within the same network?

A) Modem
B) Router
C) Switch
D) Repeater

# Section Quiz

**Question 3:**

What port number does HTTPS use

A) 21
B) 80
C) 443
D) 22

**Question 4:**

During data transfer, which of the following is used to reassemble the packets upon arriving and also to verify all packets are received.

A) Sequence Number
B) TTL
C) HOP Count
D) Checksum